

Amr Khalil

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EXPERIENCE

Senior Unity Engineer

BEDO Innovating Education

Oct 2024 - Present

6th of October, Giza

- Developed Unity-based educational and industrial simulations, including baked fluid-behavior visuals.
- Built digital twin and digital shadow systems for PLC ladder logic and PID-controlled devices.
- Collaborated with instructional design, art, and mechatronics teams to deliver simulation products.
- Improved scalability using ScriptableObject architecture and dependency injection.

Unity and AR Application Developer

Ivris

Jun 2024 - Present

Remote / Freelance

- Developed Ivris, a cross-platform Unity app for home interior visualization.
- Implemented runtime 3D model loading for furniture and room assets.
- Built Photon-based multiplayer room editing for real-time WebGL collaboration.
- Integrated Android/iOS AR previewing with PlayFab and a Render-hosted backend.

Unity Virtual Reality Developer

Tanta University

Dec 2023 - Oct 2024

Tanta, Gharbia Governorate

- Developed Meta Quest 2 VR applications for educational, lab, and medical training.
- Built a VR onboarding experience and viscosity lab simulation with interactive tools.
- Developed a multi-angle 360 surgical operation viewer with camera switching.

SELECTED PROJECTS AND AWARDS

Smart Irrigation System | IBM Call for Code 2019 Regional Winner, IoT, Machine Learning

- Built an agri-tech solution using IoT and machine learning to monitor plant health, reduce water usage, and support smarter irrigation decisions.

Nescafe Ice Coffee VR Surfing Experience | Freelance, Unity VR

- Developed a VR surfing game for a marketing booth experience promoting Nescafe Ice Coffee.

EDUCATION

German University in Cairo

Computer Science Engineering and Technology | Bachelor Grade: A | GPA: 1.79 (German Standard: B+)

Sept 2014 - July 2019

- Bachelor project: Unity-based self-driving car simulation using reinforcement learning.
- Mentored bachelor students on Unity simulation, reinforcement learning, and self-driving car project topics.
- Delivered technical sessions on Unity fundamentals, physics, NavMesh, and simulation architecture.

TECHNICAL SKILLS

Languages: C# (Unity), JavaScript (WebGL .jslib), SQL, Java

Unity Development: Unity, WebGL, SOAP, PlayMaker, ScriptableObject architecture, dependency injection

XR Development: VR, AR, Meta Quest 2, Auto Hand, 360 Video, Android, iOS

Simulation and Industrial: Digital twins, digital shadows, PLC ladder logic, TIA Portal, PID systems

Multiplayer and Backend: Photon, PlayFab, Node.js, Oracle Database, Render

Tools: Git, Visual Studio